Week 3

- **Problem 1**
  We say that two integers have *different parity* if one of them is odd and the other one is even.
  Prove that two integers have different parity if and only if their sum is odd.

  - **Proof 1.1**
    ($\Longrightarrow$) Suppose two integers have different parity. Without loss of generality, let one of them be $2k_1$ for some $k_1 \in \mathbb{Z}$ and the other one be $2k_2 + 1$ for some $k_2 \in \mathbb{Z}$. Then their sum

    $$2k_1 + (2k_2 + 1) = 2(k_1 + k_2) + 1$$

    is odd because $k_1 + k_2 \in \mathbb{Z}$.

    ($\Longleftarrow$) Let the two integers be a and b, and suppose that their sum is odd, that is, $a + b = 2s + 1$ for some $s \in \mathbb{Z}$. By definition, the integer a must be either even or odd. We consider these two cases:

    1. **Case 1**: Suppose a is even, so $a = 2k_1$ for some $k_1 \in \mathbb{Z}$.
       Then we can express b as:

       $$b = (a + b) - a = (2s + 1) - 2k_1 = 2(s - k_1) + 1.$$

       Since $s - k_1 \in \mathbb{Z}$, b is an odd integer. Thus, a is even and b is odd, meaning they have different parity.
    2. **Case 2**: Suppose a is odd, so $a = 2k_1 + 1$ for some $k_1 \in \mathbb{Z}$.
       Then we can express b as:

       $$b = (a + b) - a = (2s + 1) - (2k_1 + 1) = 2(s - k_1).$$

       Since $s - k_1 \in \mathbb{Z}$, b is an even integer. Thus, a is odd and b is even, meaning they have different parity.

    Therefore, in either case, the two integers must have different parity.

- **Problem 2**
  Suppose that n lines are drawn in the plane so that

  1. no two lines are parallel;
  2. no three lines pass through the same point.

  Prove by induction that the lines divide the plane into

  $$1 + \frac{n(n+1)}{2}$$

  regions.

  - **Proof 2.1**
    By induction.

    1. Base step
       If $n = 1$ the result is true since the number of regions is $1 + \dfrac{1 \cdot (1+1)}{2} = 2.$
    2. Inductive step
       Suppose that the result is true for n, that is, any n lines satisfying the conditions divide the plane into

       $$1 + \frac{n(n+1)}{2}$$

regions.
Now consider the case with $n+1$ lines. When we add the $(n+1)$-th line to the existing n lines:

- Since no two lines are parallel and no three lines pass through the same point, the new line must intersect all n existing lines at exactly n distinct, non-overlapping intersection points.
- These n distinct intersection points divide the new $(n+1)$-th line into exactly $n+1$ line segments or rays at the ends.
- Each of these $n+1$ segments cuts through an existing region and splits it into two, thereby creating exactly $n+1$ new regions.

Hence, the total number of regions for $n+1$ lines is

$$\begin{aligned}\left(1+\frac{n(n+1)}{2}\right)+(n+1) &= 1+\frac{n(n+1)+2(n+1)}{2} \\ &= 1+\frac{(n+1)(n+2)}{2} \\ &= 1+\frac{(n+1)((n+1)+1)}{2}.\end{aligned}$$

Then the result is true for $n+1$ and therefore the lines divide the plane into $1 + \dfrac{n(n+1)}{2}$ regions for every $n \in \mathbb{N}$.

- **Problem 3**
Prove by induction that for every $n \in \mathbb{N}_0$,

$$4 \mid \left(5^{n+1}+3\cdot 9^n-4\right).$$

  - **Proof 3.1**
  By induction.

    1. Base step
    If $n=0$ then the result is true because $5^{0+1}+3\cdot 9^0-4=4$ and $4 \mid 4$.
    2. Inductive step
    Suppose that the result is true for n, that is,

    $$4 \mid \left(5^{n+1}+3\cdot 9^n-4\right).$$

    By definition, $\exists k \in \mathbb{Z}, 5^{n+1}+3\cdot 9^n-4=4k \implies 5^{n+1}+3\cdot 9^n=4(k+1)$. Then

    $$\begin{aligned}5^{(n+1)+1}+3\cdot 9^{n+1}-4 &= 5\cdot 5^{n+1}+9\cdot 3\cdot 9^n-4 \\ &= 5\cdot\left(5^{n+1}+3\cdot 9^n\right)+4\cdot 3\cdot 9^n-4 \\ &= 5\cdot 4(k+1)+4\cdot 3\cdot 9^n-4 \\ &= 4\cdot\left(5k+4+3\cdot 9^n\right).\end{aligned}$$

    Since $5k+4+3\cdot 9^n \in \mathbb{Z}$, we have $4 \mid \left(5^{(n+1)+1}+3\cdot 9^{n+1}-4\right)$. Then the result is true for $n+1$ and therefore

    $$4 \mid \left(5^{n+1}+3\cdot 9^n-4\right)$$

    is true for every $n \in \mathbb{N}_0$.

- **Problem 4**

Suppose $a, c \in \mathbb{N}$ with

$$a < c \quad \text{and} \quad a \mid c.$$

Suppose also that

$$c - a \mid c + a.$$

Determine all possible values of c in terms of a. Justify your answer.
Your proof should use only the definition and basic properties of divisibility.

- **Solution 4.1**
  Suppose $a, c \in \mathbb{N}$ with $a < c, a \mid c, c - a \mid c + a$. The only possible values of c in terms of a are $c = 2a$ and $c = 3a$.
  By the basic properties of divisibility, since $c - a \mid c + a$ and $c - a \mid c - a$, it must divide their difference:

  $$c - a \mid (c + a) - (c - a) \implies c - a \mid 2a.$$

  Since $a \mid c$ and $a < c$, by the definition of divisibility, there exists an integer $k \in \mathbb{N}$ with $k > 1$ such that

  $$c = ka.$$

  Substituting $c = ka$ into the relation $c - a \mid 2a$, we get

  $$ka - a \mid 2a \implies a(k-1) \mid 2a.$$

  Since $a \in \mathbb{N}$, by the basic properties of divisibility

  $$k - 1 \mid 2.$$

  Since $k > 1$, the term $k - 1$ is a positive integer. The only positive divisors of 2 are 1 and 2. This gives two possible cases: $k = 2$ or $k = 3$, that is, $c = 2a$ or $c = 3a$.

- **Problem 5**
  Suppose $x, y \in \mathbb{N}$ with

  $$x^6 = 81y^{10}.$$

  Prove that every prime that divides y also divides x.

  - **Proof 5.1**
    Suppose $x, y \in \mathbb{N}$ with $x^6 = 81y^{10}$. Let p be a prime number such that $p \mid y$. Since $y \in \mathbb{N}$, by definition, we have $\mho_p(y) \geqslant 1$.
    By Proposition, we have

    $$\begin{aligned} \mho_p\left(x^6\right) &= \mho_p\left(81y^{10}\right) \\ 6\mho_p(x) &= \mho_p(81) + 10\mho_p(y). \end{aligned}$$

    Since $81 = 3^4$, we know that $\mho_p(81) \geqslant 0$ for any prime p. Together with $\mho_p(y) \geqslant 1$, we obtain that

    $$6\mho_p(x) = \mho_p(81) + 10\mho_p(y) \geqslant 0 + 10(1) = 10.$$

    This implies $6\mho_p(x) \geqslant 10$, which means $\mho_p(x) \geqslant \dfrac{10}{6} > 0$.
    Since $\mho_p(x) \in \mathbb{N}_0$, it must be that $\mho_p(x) \geqslant 1$. By definition, this means $p \mid x$.
    Therefore, every prime that divides y also divides x.